In this topic we design and implement a database level loose coupling solution that can propagate events from an information system into another. The objective is that the process should not at the least interfere with the source system. It cannot significantly slow it down, nor may any error that may occur while processing an event on the target side be propagated back to the source system.

For this reason, we use the loose coupling concept. We only log the INSERT and UPDATE events on the source table via a trigger, and process these events in batches executed by a scheduled job. The job also keeps track of the state and results of the processing of each event. Since the processing of the event is performed out-of-process, a processing error does not manifest as an error in the source system.

The topic is presented as a case study in a hypothetical (but realistic) trading company using the Northwind database.

After completing this topic

- you will understand the loose vs tight coupling concept
- you will be able to design and work with common DML triggers
- you will be able to use cursors and stored procedures
- you will be able to design, start and manage database jobs